Octahedral Proof Calculus: A Geometric Meta-Language for Distinction

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Date: March 2026
Series: Distinction Observable Theory (DOT), Supplementary Publication
arXiv subject: Mathematical Physics (math-ph); Logic in Computer Science (cs.LO)

> "The unmanifest [Prakriti, primal matter] is endowed with the three gunas… The gunas, by their nature, are pleasure [Sattva], pain [Rajas], and indifference [Tamas]. Their purpose is to illuminate, to activate, and to restrict. They suppress each other, support each other, produce each other, and act together. Sattva (light) is considered light and illuminating; Rajas (passion) is stimulating and mobile; Tamas (darkness) is heavy and concealing. They work together for a single purpose, like a lamp [where oil, wick, and flame act in concert].
>
> …
>
> Purusha [pure consciousness] is merely a witness to what occurs. It is endowed with absolute isolation, complete indifference, the capacity to contemplate, and absolute non-action.
>
> …
>
> For the sake of Purusha's contemplation of Prakriti, and for its ultimate liberation, their union occurs. It is like the union of the lame and the blind. From this union, creation arises."
> — *Sankhya Karika, 11–21*

> "In the beginning was the Word, and the Word was with God, and the Word was God. He was in the beginning with God. All things were made through Him, and without Him nothing was made that was made." — *Gospel of John, 1:1–3*

Abstract

We introduce the **Octahedral Proof Calculus** (OPC) — a geometric meta-language in which logical formulas are topological subsets of the complete tripartite graph $K(2,2,2)$ (the octahedron), logical transformations are geometric flows driven by the nilpotent boundary operator ∂ ($\partial^2 = 0$), and proof complexity is measured by a structural defect $\mathcal{D}$ isomorphic to the particle-physics defect sectors of the Distinction Observable Theory (DOT; i.e. a theory of observable distinctions). The calculus is built from exactly **six primitive operations** — SELECT (σ), NEGATE ($\bar{\sigma}$), SPAN (ε), FILL (φ), FLOW (∂), and ROTATE (ρ) — which constitute a complete instruction set for the octahedral machine. We show that (i) the eight faces of the octahedron are isomorphic to the eight rows of a three-variable truth table, making OPC a natural geometric encoding of propositional logic; (ii) arbitrary n-variable logic is handled by fractal nesting (tensor products of octahedra),

where one nesting step yields exactly 64 states — the vacuum configuration count of DOT; (iii) every physical quantity (velocity, force, energy, action) possesses an absolute octahedral address derived from its dimensional exponents (l, t, m) over the three axes Length, Time, Mass; (iv) the Heisenberg uncertainty principle emerges as a Borromean prohibition on simultaneous readout of a pole and its conjugate edge within a single face. As a demonstration, we reformulate the **Yang–Mills mass gap** proof entirely in OPC notation: a five-step proof-flow ($\sigma \to \varepsilon^3 \to \partial \to \varphi \to \rho$) with defect $\mathcal{D} = 1/3$, showing that the mass gap is an elementary spectral fact of the discrete carrier rather than a deep mystery of the continuum.

Keywords: proof calculus, octahedral meta-language, boundary operator, nilpotency, physical dimension addressing, Yang–Mills mass gap, $K(2, 2, 2)$ graph, Borromean topology, Turing completeness.

This is a supplementary publication to the three-part DOT series. Part I [arXiv:XXXX.XXXXX] establishes the carrier. Part II [arXiv:XXXX.XXXXX] derives mathematics. Part III [arXiv:XXXX.XXXXX] reads out physics.

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§1. Introduction: Why a Geometric Proof Language?

Mathematical proofs are traditionally written as linear sequences of symbols. But the underlying structure of a proof is not linear — it is **topological**. A proof has branching, merging, symmetry, and closure. These properties are invisible in text but manifest in geometry.

The Distinction Observable Theory (DOT) has shown that a single finite graph — the complete tripartite graph $K(2, 2, 2)$, an octahedron with 6 poles, 12 edges, and 8 faces — generates all of mathematics (Part II) and all of physics (Part III) through the action of a single nilpotent boundary operator ∂ ($\partial^2 = 0$). The natural question arises: can the **process of proof itself** be projected onto the same carrier?

This paper answers affirmatively. We construct the **Octahedral Proof Calculus** (OPC), a geometric meta-language where:

1. **Formulas** are subsets of the 8 faces of the octahedron.
2. **Proof steps** are geometric operations (flows, rotations, intersections).
3. **Proof complexity** is measured by structural defect $\mathcal{D}$, isomorphic to the particle defect sectors of DOT.
4. **Physical quantities** (velocity, energy, force) have absolute octahedral addresses.

The result is a three-way isomorphism:

$$\text{Logic} \leftrightarrow \text{Topology} \leftrightarrow \text{Physics}$$

All three domains are described by **the same geometric machine**: $K(2, 2, 2)$ with operator ∂.

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§2. The Octahedron as Logical Space

2.1. Three Axes, Eight Faces

The octahedron $K(2,2,2)$ possesses:

Element	Count	Logical role
Axes	3	Boolean variables A, B, C
Poles	6 ($= 2 \times 3$)	Literals: $A, \neg A, B, \neg B, C, \neg C$
Edges	12	Pairwise conjunctions ($A \wedge B$, $A \wedge \neg C$, etc.)
Faces	8 ($= 2^3$)	Complete truth assignments (minterms)

Each face is a triangle connecting three poles from distinct axes. The face $(+X, -Y, +Z)$ corresponds to the truth assignment $A = \text{True}$,$B = \text{False}$,$C = \text{True}$.

Theorem 2.1. *The 8 faces of $K(2,2,2)$ are in bijection with the $2^3 = 8$ rows of the truth table for three Boolean variables.*

Proof. Each face selects exactly one pole from each of the three axes. The choice of $+$ or $-$ on axis i corresponds to the truth value of variable i. Since there are 2^3 such selections, and each selection uniquely determines a face, the bijection is exact. ▫

2.2. Formulas as Face Subsets

Any propositional formula $\Phi(A, B, C)$ defines a subset $S_\Phi \subseteq \{f_1, ..., f_8\}$ of the eight faces — namely, those faces (truth assignments) that make Φ true.

Formula	$\lvert S_\Phi \rvert$	Geometry
False	0	Empty set (no faces)
A	4	Four faces containing $+X$
$A \wedge B$	2	Two faces containing both $+X$ and $+Y$
$A \Longrightarrow B$	6	Eight faces minus two containing $+X$ and $-Y$
$A \oplus B$ (XOR)	4	Alternating faces
True	8	Full octahedron

Observation. A tautology ($|S_\Phi| = 8$) is geometrically the **complete octahedron**. A contradiction ($|S_\Phi| = 0$) is the **empty set**. All intermediate formulas are partial octahedra — geometric objects between void and wholeness.

2.3. Logical Operations as Geometric Operations

Logical operation	Geometric operation on face subsets
$\neg\Phi$	Complement: $S_{\neg\Phi} = \{f_1, ..., f_8\} \setminus S_\Phi$
$\Phi_1 \wedge \Phi_2$	Intersection: $S_{\Phi_1 \wedge \Phi_2} = S_{\Phi_1} \cap S_{\Phi_2}$
$\Phi_1 \vee \Phi_2$	Union: $S_{\Phi_1 \vee \Phi_2} = S_{\Phi_1} \cup S_{\Phi_2}$
$\Phi_1 \Longrightarrow \Phi_2$	$S_{\neg\Phi_1} \cup S_{\Phi_2}$
$\Phi_1 \oplus \Phi_2$	Symmetric difference: $S_{\Phi_1} \triangle S_{\Phi_2}$

Theorem 2.2 (Modus Ponens as Intersection). *Given S_A (4 faces around $+X$) and $S_{A \Longrightarrow B}$ (6 faces), their intersection $S_A \cap S_{A \Longrightarrow B}$ yields exactly the 4 faces around $+Y$ — i.e., S_B. Thus B is geometrically inevitable.*

Proof. $S_{A \Longrightarrow B}$ consists of all faces except those with $+X$ and $-Y$. S_A consists of all faces with $+X$. Their intersection keeps only faces with $+X$ **and** $+Y$ (since $-Y$ faces were removed). These are exactly the faces of $S_{A \wedge B}$, which is a subset of S_B. More precisely: $|S_A \cap S_{A \Longrightarrow B}| = |S_A| - |S_A \cap S_{\neg(A \Longrightarrow B)}| = 4 - 2 = 2$. These 2 faces lie on $+Y$. By axiom completeness, if A is given (all $+X$ faces are active) and $A \Longrightarrow B$ is given, the only surviving faces are those with $+Y$ active. Hence B holds. ▫

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§3. Alphabet of Operations: The Six Primitives

The octahedron $K(2, 2, 2)$ possesses exactly **three types of elements** (poles, edges, faces) and **two directions** of the boundary operator ∂ ($\partial_\uparrow$: faces $\longrightarrow$ edges; $\partial_\downarrow$: edges $\longrightarrow$ poles). From this, exactly **six primitive operations** emerge — one per pole:

#	Operation	Geometry on $K(2,2,2)$	Logical analog	Symbol
1	**SELECT**	Activate a pole on an axis	Assert a variable	σ
2	**NEGATE**	Switch to the antipodal pole	Logical $\neg$	$\bar{\sigma}$
3	**SPAN**	Stretch an edge between two poles of different axes	Conjunction $\wedge$	ε
4	**FILL**	Form a face (triangle) from three edges	Complete state (minterm)	φ
5	**FLOW**	Run ∂ along face $\longrightarrow$ edge $\longrightarrow$ pole	Modus Ponens / deduction	∂
6	**ROTATE**	Apply an element of the symmetry group $S_3 \times \mathbb{Z}_2^3$	Variable substitution	ρ

Any proof is a finite sequence of these six symbols. This is the "machine code" of the Octahedron.

3.1. Composition Laws

The operations obey strict algebraic rules:

1. $\sigma \circ \bar{\sigma} = \emptyset$ — selecting a pole and its antipode annihilate each other (Borromean prohibition: no "monopole").
2. $\varepsilon \circ \varepsilon \circ \varepsilon = \varphi$ — three edges close into a face. Three conjunctions create a complete state.
3. $\partial \circ \partial = 0$ — **nilpotency**. A double flow always annihilates. A proof applied twice yields no new knowledge.
4. $\rho^{48} = \mathrm{id}$ — the full rotation group ($|O_h| = 48$) closes.

3.2. Formal Grammar

An OPC proof is a sequence $\Pi = (o_1, o_2, ..., o_n)$ where each $o_i \in \{\sigma, \bar{\sigma}, \varepsilon, \varphi, \partial, \rho\}$, subject to:

- **Well-formedness:** ε requires two previously activated poles on distinct axes.
- **Closure:** φ requires three previously formed edges sharing three distinct poles.
- **Termination:** The proof concludes when either (a) ∂-flow reaches the target pole (theorem proved), (b) $\sigma \circ \bar{\sigma} = \emptyset$ collapse occurs (proof by contradiction), or (c) flow stalls (undecidable in current axioms).

§4. Fractal Scaling: n Variables

A single octahedron accommodates 3 variables. Arbitrary n is handled by two mechanisms.

4.1. Tensor Octahedron

For $n = 3k$ variables, construct the **tensor product** of k octahedra:

$$\mathcal{O}^{(k)} = \underbrace{K(2,2,2) \otimes K(2,2,2) \otimes \cdots}_{k \text{ times}}$$

- Number of faces: $8^k = 2^{3k} = 2^n$ — matches the full truth table.
- Number of poles: $6k = 2n$.
- Number of edges: $12k$ + inter-octahedral links.

4.2. Nested Octahedron (Recursion)

An alternative approach: each pole of the first octahedron **becomes** a new octahedron.

- **Level 0:** 1 octahedron, 8 faces. Encodes $2^3 = 8$ states.
- **Level 1:** Each of 6 poles unfolds into an octahedron. Total: $8 \times 8 = 64$ states — **exactly the vacuum configuration count** of DOT!
- **Level 2:** $64 \times 8 = 512$ states. Sufficient for 9 variables.

Key observation: The theory confirms itself. One nesting step ($8 \to 64$) yields exactly **64 vacuum configurations** — the same number from which DOT constructs all of particle physics. Logic and physics are the same structure.

4.3. Partial Octahedra ($n \neq 3k$)

If n is not a multiple of 3, the last octahedron is used partially:

n	Octahedra	Last octahedron
1	1	Only 1 axis (2 poles, 0 edges)
2	1	2 axes (4 poles, 4 edges)
3	1	Full
4	2	1st full + 2nd: 1 axis
7	3	2 full + 3rd: 1 axis

§5. Defect as Proof Complexity

In DOT, every physical particle carries a structural **defect** $\mathcal{D}$. Analogously, every proof-flow carries its own defect — a measure of topological complexity.

5.1. Definition

$$\mathcal{D}_{\text{proof}} = 1 - \frac{|\text{faces used by the flow}|}{|\text{faces available after axioms}|}$$

- $\mathcal{D} = 0$: Trivial proof (tautology). Flow covers all available space. Analog of **massless** states (photon).
- $\mathcal{D} = 1/3$: Flow covers 2/3 of available space. One axis is "defective." Analog of **quarks** — partial, fragmented verification.
- $\mathcal{D} = 1$: Full dispersal. Every proof step distinguishes all three axes. Analog of **leptons** — complete but laborious work.
- $\mathcal{D} = 4/3$: Proof by contradiction. Super-stressed flow requiring structural collapse. Analog of **heavy bosons**.

5.2. Hierarchy of Proof Types

Proof class	$\mathcal{D}$	Characteristic	Physical analog
Tautology	0	Instantaneous, effortless	Massless (γ, gluons)
Direct deduction	1/3	Partial axis usage	Quarks (confinement: steps are bound)
Induction	1	Full deployment across all axes	Leptons (free particle)
Proof by contradiction	4/3	Requires "explosion" — assuming false, then collapse	W, Z (breaking all symmetries)

Corollary (Gödel's Incompleteness). A statement with $\mathcal{D} > 4/3$ exceeds the horizon of the octahedron. Its defect cannot be closed within a single level — a meta-transition to the next fractal level is required. Gödel sentences are **trans-horizonal proofs**.

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§6. Absolute Addressing of Physical Quantities

6.1. Three Axes as Dimensions

Every physical quantity has a **dimension** — a combination of three fundamental units: Length (L), Time (T), and Mass (M). These are exactly the three axes of the Octahedron.

Octahedron axis	Fundamental dimension	Pole +	Pole −
X	**Length (L)**	Spatial extension	Point localization
Y	**Time (T)**	Duration / process	Instantaneity
Z	**Mass (M)**	Inertia / energy weight	Masslessness

Each pole carries a **power exponent** ($+n$ or $-n$), corresponding to the dimensional exponent. For example, velocity $[v] = LT^{-1}$ means: axis X with exponent $+1$, axis Y with exponent -1, axis Z with exponent 0.

6.2. Address Table of Physical Quantities

Each physical quantity receives an **absolute address** — a triple (l, t, m) of dimensional exponents, uniquely encoding it as a face, edge, or pole of the Octahedron.

Layer $\Omega_{0/1}$ — Dimensionless Constants (Center $\vec{0}$)

Quantity	Dimension	Address (l, t, m)	Position on octahedron
Dimensionless constant (α, π, γ)	1	$(0, 0, 0)$	**Center** — vacuum point

Dimensionless constants are **scalars** living at the center. They do not "pull" any axis. $\alpha^{-1} \approx 137.036$ is a pure number, address $(0, 0, 0)$.

Layer $\Omega_{1/2}$ — One-Dimensional (Poles)

Quantity	Dimension	Address	Pole
Length (x)	L	$(1, 0, 0)$	$+X$
Time (t)	T	$(0, 1, 0)$	$+Y$
Mass (m)	M	$(0, 0, 1)$	$+Z$

The three fundamental quantities are exactly **three poles** of the Octahedron.

Layer $\Omega_{2/3}$ — Two- and Three-Dimensional (Edges and Faces)

Quantity	Dimension	Address	Element
Velocity (v)	LT^{-1}	$(1,-1,0)$	Edge $+X \leftrightarrow -Y$
Frequency (ν)	T^{-1}	$(0,-1,0)$	Antipole $-Y$
Momentum (p)	MLT^{-1}	$(1,-1,1)$	**Face** $(+X,-Y,+Z)$
Acceleration (a)	LT^{-2}	$(1,-2,0)$	Fractal step: $+X \leftrightarrow (-Y)^2$
Force (F)	MLT^{-2}	$(1,-2,1)$	Face on 2nd fractal level
Energy (E)	ML^2T^{-2}	$(2,-2,1)$	2nd fractal level
Pressure (P)	$ML^{-1}T^{-2}$	$(-1,-2,1)$	Antipodal-fractal face
Power (W)	ML^2T^{-3}	$(2,-3,1)$	3rd fractal level
Action (S)	ML^2T^{-1}	$(2,-1,1)$	Face $(+X^2,-Y,+Z)$

6.3. Key Observations

1. Momentum is a face of the Octahedron. $p = mv$, dimension MLT^{-1}, address $(1,-1,1)$. All three axes are engaged — this is a complete **face** (triangle $+X,-Y,+Z$). Momentum is the minimal complete physical object.

2. Energy requires a fractal step. $E = mc^2$, dimension ML^2T^{-2}. The exponent L^2 means axis X is activated twice — a **second octahedron** in the nested structure is needed. Energy lives **on the 2nd fractal level**. This explains why energy is a less "fundamental" quantity than momentum: it requires a greater depth of distinction.

3. Action $S = Et$ and the uncertainty principle. Address $(2,-1,1)$. The quantum of action $\hbar$ is the minimal face area on the 2nd level. The Heisenberg uncertainty principle ($\Delta x \cdot \Delta p \geq \hbar/2$) is a **Borromean prohibition** on simultaneous exact readout of a pole and its conjugate edge within a single face.

4. Force is a dimensional defect. $F = ma$, address $(1,-2,1)$. The exponent T^{-2} (double negation of time) creates an "overstress" on axis Y. Force is not an object but a measure of inter-axial stress — a **defect gradient** in geometrical language.

5. Pressure is antipodal force. $P = F/A$, address $(-1,-2,1)$. Axis X is inverted (L^{-1}). Pressure is force "turned inside out": instead of extension, compression.

6.4. Field Quantities

Quantity	Dimension	Address	Layer Ω	DOT meaning
Electric charge (q)	$[q]$	Axial mode	$\Omega_{2/3}$	Projection of dichotomy onto 1 axis of 3
Electric field (E)	$MLT^{-3}A^{-1}$	3rd level	$\Omega_{3/4}$	Gradient ∇ of potential
Magnetic field (B)	$MT^{-2}A^{-1}$	3rd level	$\Omega_{3/4}$	Curl $\nabla\times$ of potential
Yang–Mills field strength ($F^a_{\mu\nu}$)	Curvature tensor	Multi-level	$\Omega_{4/5}$	Connection curvature on $K(2,2,2)$
Coupling constant (g_s)	Dimensionless	$(0,0,0)$ at μ_0	$\Omega_{5/6}$	Scale-dependent scalar (beta function)

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§7. Worked Example: Irrationality of $\sqrt{2}$

A classical proof by contradiction. How does it appear in OPC?

7.1. Octahedron Setup

Assign axes: - **Axis X:** Parity ($+X$ = numerator p is even, $-X$ = odd) - **Axis Y:** Divisibility ($+Y$ = p^2 divisible by 2, $-Y$ = not) - **Axis Z:** Irreducibility ($+Z$ = fraction p/q is irreducible, $-Z$ = reducible)

7.2. Step 1 — Axiomatics (Sculpting)

Assumption: $\sqrt{2} = p/q$, fraction irreducible. - $\sigma(+Z)$ — activate the irreducibility pole. - Borromean restriction: with $+Z$ active, faces containing $-Z$ are forbidden. From 8 faces, 4 remain.

7.3. Step 2 — Flow (Algebraic Unfolding)

$p^2 = 2q^2 \Longrightarrow p^2$ even $\Longrightarrow p$ even. - ∂-flow: $+Z \to \varepsilon(+Z,+Y) \to \sigma(+Y) \to \varepsilon(+Y,+X) \to \sigma(+X)$. - Result: poles $+X, +Y, +Z$ all active. This is the single face $(+X,+Y,+Z)$.

7.4. Step 3 — Second Flow (Substitution $p = 2m$)

$p = 2m \Longrightarrow 4m^2 = 2q^2 \Longrightarrow q^2$ even $\Longrightarrow q$ even. - Substitution = ρ (rotation: $X \leftrightarrow Y$). - Flow yields: $+X$ is active for q as well. Both p and q are even.

7.5. Step 4 — Collapse (Defect Readout)

- Both even $\Longrightarrow$ fraction is reducible: $\sigma(-Z)$.
- But Step 1 fixed $\sigma(+Z)$.
- $\sigma(+Z) \wedge \sigma(-Z) = \emptyset \longrightarrow$ **Borromean annihilation!**

The octahedron collapses. The assumption is impossible. $\sqrt{2}$ is irrational. ∎

Proof defect: $\mathcal{D} = 4/3$ (proof by contradiction, all 3 axes engaged + forced collapse).

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§8. The Yang–Mills Mass Gap in OPC Notation

8.1. Statement of the Problem

The official Clay Institute formulation (2000):

> *For any compact simple gauge group G, prove that the quantum Yang–Mills theory on $\mathbb{R}^4$ exists and has a mass gap $\Delta > 0$: the lowest energy excitation of the vacuum is strictly positive.*

8.2. Octahedron Setup

Axes: - **X** = Spatial extension (L) - **Y** = Temporal duration (T) - **Z** = Inertia / mass (M)

Axioms (Sculpting): 1. $\partial^2 = 0$ — nilpotency (Borromean prohibition). 2. Laplacian spectrum: $\sigma(L) = \{0, 4, 4, 4, 6, 6\} \longrightarrow$ spectral gap $\Delta_{\text{spec}} = 4$. 3. Gauge group $SU(3)$ = 8 faces of the Octahedron (color octet of gluons).

8.3. Five-Step Proof-Flow

Step 1. SELECT — What is an "observable state"?

$$\sigma(+Z): \quad \text{Activate the mass pole.}$$

An observable physical state is a closed contour on the graph: $\partial\psi = 0$. It must have address (l, t, m) with $m \geq 0$. The operator $\sigma(+Z)$ selects: we seek states possessing mass.

Step 2. SPAN — Minimal closed contour.

$$\varepsilon(+X, +Z): \quad \text{Stretch the edge "extension} \times \text{mass".}$$

$$\varepsilon(-Y, +Z): \quad \text{Stretch the edge "time-inversion} \times \text{mass".}$$

$$\varepsilon(+X, -Y): \quad \text{Stretch the edge "velocity" } (\ LT^{-1}\ ).$$

Three edges close: $\varepsilon \circ \varepsilon \circ \varepsilon = \varphi$. Result: a face — the **minimal closed contour** on the Octahedron. This is momentum: address $(1, -1, 1)$.

Step 3. FLOW — Spectral constraint.

$$\partial(\varphi) \to \text{spectral projection.}$$

The operator ∂, acting on the face, projects it onto spectral subspaces of the Laplacian. The face must excite mode $\lambda \geq \lambda_1 = 4$ (non-zero). Between $\lambda_0 = 0$ (vacuum) and $\lambda_1 = 4$ **there are no intermediate modes**. The ∂-flow cannot "leak" toward zero continuously — it must "jump" across the gap.

Step 4. FILL — Defect quantum.

The minimal structural defect on $K(2,2,2)$ is $\mathcal{D}_{\min} = 1/3$ (§26–27 of Part III). A face excited on mode λ_1 carries defect $\mathcal{D} \geq 1/3 > 0$. By the mass lever formula:

$$M = \frac{C}{\gamma^k} \cdot m_e, \quad C \geq 1, \quad \gamma < 1 \Longrightarrow M > 0.$$

The face $\varphi(+X, -Y, +Z)$ possesses strictly positive mass.

Step 5. ROTATE — Confinement.

$$\rho: \quad \text{Can an edge be "torn" from a face?}$$

Attempting to isolate edge $(+X, -Y)$ (a quark) from face $(+X, -Y, +Z)$ requires $\partial(\text{edge}) \neq 0$ — the edge is not closed, it is not an independent observable. By axiom $\partial^2 = 0$: the boundary of a face ($\partial\varphi$) = sum of edges, but the boundary of an edge ($\partial\varepsilon$) = difference of poles $\neq 0$. A quark is an unclosed boundary. A free quark produces a contradiction: $\partial\psi = 0 \wedge \partial\psi \neq 0$. **Borromean annihilation!**

8.4. Result Readout

Step	Operation	Result
1	σ	Massive state possible
2	$\varepsilon^3 = \varphi$	Minimal contour = momentum
3	∂	Spectral gap $\lambda_1 = 4$ forbids $M \to 0$
4	φ	$M \geq C \cdot m_e/\gamma^k > 0$
5	ρ	Quark = unclosed boundary $\longrightarrow$ confinement

Proof defect: $\mathcal{D} = 1/3$. This is a direct deductive proof (not by contradiction), but with fragmented axis usage (mass is fully engaged, time only partially through the spectral structure). Analog: the **quark sector** — the proof itself is "bound," like quarks in a hadron.

8.5. Why the Standard Approach Does Not Work

In the standard formulation (Clay Institute), one must prove the mass gap on the **continuous** space $\mathbb{R}^4$. But on $\mathbb{R}^4$ the Laplacian has a **continuous** spectrum $[0, \infty)$ — there is no gap by definition.

In OPC, this is transparent: the continuous limit ($\mathbb{R}^4$) is the limit of infinite fractal nesting ($k \to \infty$). As $k \to \infty$ the spectral gap **is preserved** at every level (a property of tensor products — the gap of the largest factor is inherited), but its absolute magnitude in observer units scales as $\Delta(k) = 4 \cdot \gamma^k$. In the continuous limit the gap, expressed in "continuous" units, becomes infinitesimally small — but **strictly positive** for any finite k.

The paradox of the Clay formulation: it demands a gap in a system where a gap is impossible by construction (continuous spectrum). DOT shows the question is ill-posed: the correct formulation is "does there exist a discrete carrier from which Yang–Mills QFT emanates with gap preservation?" Answer: yes, it is $K(2,2,2)$.

§9. Connections to Models of Computation

The OPC meta-language is isomorphic to several fundamental structures in computer science:

Model	Isomorphism with the Octahedron
Boolean circuit	Each face = gate, edges = wires, ∂ = signal propagation
Simplicial complex	Octahedron = 2-dimensional simplicial complex, ∂ = standard boundary operator
Category	Poles = objects, edges = morphisms, $\partial^2 = 0$ = associativity
Lambda calculus	σ = abstraction (variable binding), ∂ = β-reduction
Cellular automaton	8 faces = 8 cells, ∂ = transition rule, $\partial^2 = 0$ = prohibition of superluminal propagation

Key consequence: Since OPC is Turing-complete (through fractal scaling it models arbitrary Boolean circuits), $\boldsymbol{\partial^2 = 0}$ **is the minimal axiom of computation**. A single nilpotency rule suffices to generate all computability.

§10. Summary and Outlook

The Octahedral Proof Calculus (OPC) is not a metaphor. It is a strict isomorphism between:

- **Logic** (formulas, inference rules)
- **Topology** (simplicial complexes, boundary operator)
- **Physics** (defect, mass, confinement)

Three subject domains are described by **one and the same geometric machine** $-$ $K(2,2,2)$ with operator ∂. This is the strongest argument in favor of DOT as a foundational theory: it does not describe the world, it **is** the language in which the world describes itself.

Concrete next steps:

1. **Formal verification:** Implement the 6 primitives as a software module (Python / Lean4) and automatically verify proofs.
2. **Theorem catalog:** Translate 10–20 classical theorems (Pythagoras, infinitude of primes, FLT for $n = 4$) into OPC notation.
3. **Visualizer:** 3D animation of proof-flow on WebGL.
4. **Publication:** Format as a fourth DOT volume ("Part IV: Metalanguage") or as a standalone article "Octahedral Proof Calculus."

Distinction Observable Theory — Octahedral Proof Calculus
First edition, March 22, 2026